POSTS AND TELECOMMUNICATIONS INSTITUTE OF TECHNOLOGY

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Chapter 1: The limit and continuity of functions of one variable

CALCULUS 1

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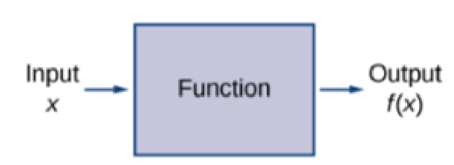
Content

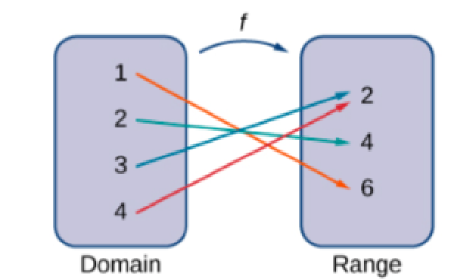
Definition

A **function** f consists of a set of inputs, a set of outputs, and a rule for assigning each input to exactly one output. The set of inputs is called the **domain** of the function. The set of outputs is called the **range** of the function.



1.1.1. Definition

- For a general function f with domain D, we often use x to denote the input and y to denote the output associated with x. When doing so, we refer to x as the independent variable and y as the dependent variable, because it depends on x. Using function notation, we write $y = f(x)$, and we read this equation as " y equals f of x."
- For the squaring function, we write $f(x) = x^2$



Definition

Let $X \subseteq \mathbb{R}, Y \subseteq \mathbb{R}$, the map

$$\begin{aligned} f : X &\to Y \\ x &\mapsto y = f(x) \end{aligned}$$

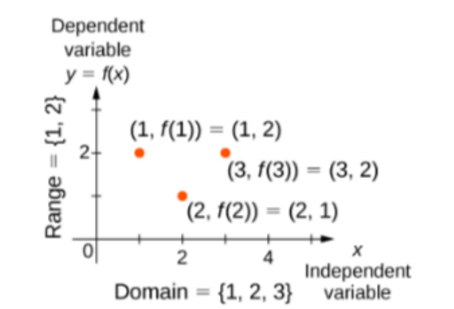
is called a **function of one variable**, the set X is called the *domain* of the function f. Normally, x is called the argument, y is the function. The value set of the function $y = f(x)$ is a set of the form $f_{val} = \{f(x) : \ x \in X\}$.

The graph

The **graph** of a function is the set of all points (x, y) in the coordinate plane where $y = f(x)$.

Example

Consider the function f, where the domain is the set $D = \{1, 2, 3\}$ and the rule is $f(x) = 3 - x$ in the following figure.



Hướng dẫn

Với mọi $x \in (0, \frac{\pi}{2}]$, ta có

$$f(x) = \frac{\ln(\sin x)}{\sqrt{x}},$$

Từ $x \in (0, \frac{\pi}{2}]$, suy ra $\sin x \in (0, 1]$ và $\ln(\sin x) \in (-\infty, 0]$,

Example

For each of the following functions, determine the domain and the range.

a. $f(x) = (x-4)^2 + 5$

b. $f(x) = \sqrt{3x+2} - 1$

c. $f(x) = \dfrac{3}{x-2}$

Definition

We say that a function f is **increasing on the interval** I if for all $x_1, x_2 \in I$,

$$f(x_1) \leq f(x_2) \text{ when } x_1 < x_2$$

We say f is **strictly increasing on the interval** I if for all $x_1, x_2 \in I$,

$$f(x_1) < f(x_2) \text{ when } x_1 < x_2$$

We say that a function f is **decreasing on the interval** I if for all $x_1, x_2 \in I$

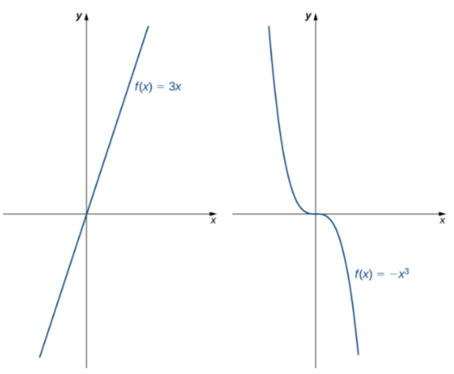
$$f(x_1) \geq f(x_2) \text{ if } x_1 < x_2$$

We say that a function f is **strictly decreasing on the interval** I if for all $x_1, x_2 \in I$

$$f(x_1) > f(x_2) \text{ if } x_1 < x_2$$

Example

The function $f(x) = 3x$ is increasing on the interval $(-\infty, \infty)$ because $3x_1 < 3x_2$ whenever $x_1 < x_2$. On the other hand, the function $f(x) = -x^3$ is decreasing on the interval $(-\infty, \infty)$ because $-x_1^3 > -x_2^3$ whenever $x_1 < x_2$



Definition

- The function $y = f(x)$ is called **upper bounded** on the domain D if and only if there exists a constant $M \in \mathbb{R}$ such that

$$f(x) \leq M \quad \forall x \in D.$$

- The function $y = f(x)$ is called **lower bounded** on the domain D if and only if there exists a constant $m \in \mathbb{R}$ such that

$$f(x) \geq m \quad \forall x \in D.$$

- The function $y = f(x)$ is called **bounded** on the domain D if and only if f is bounded above and lower bounded on D.

Infimum and supremum

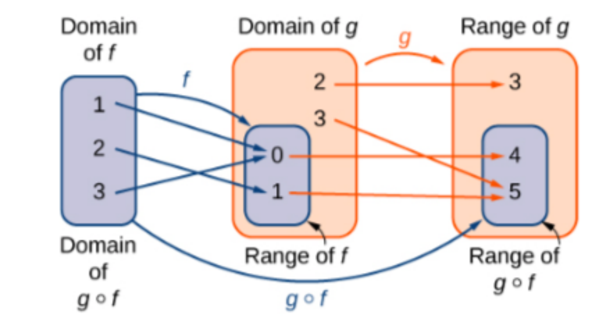
Definition

- The number m is called **infimum** of the function f on the domain D, denoted by $m = \inf_{x \in D} f(x)$ if and only if m is the largest number that satisfies
$$f(x) \geq m \quad \forall x \in D.$$
- The number M is called **supremum** of the function f on the domain D, denoted
$$M = \sup_{x \in D} f(x)$$
if and only if M is the smallest number satisfying
$$f(x) \leq M \quad \forall x \in D.$$

Composite functions

Consider the function f with domain A and range B, and the function g with domain D and range E. If B is a subset of D, then the **composite function** $(g \circ f)(x)$ is the function with domain A such that

$$(g \circ f)(x) = g(f(x))$$

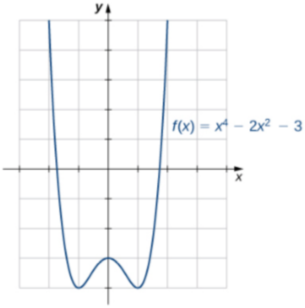


Example

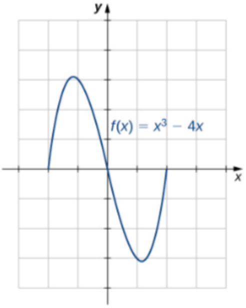
Consider the functions $f(x) = x^2 + 1$ and $g(x) = 1/x$.
a. Find $(g \circ f)(x)$ and state its domain and range.
b. Evaluate $(g \circ f)(4), (g \circ f)(-1/2)$.
c. Find $(f \circ g)(x)$ and state its domain and range.
d. Evaluate $(f \circ g)(4), (f \circ g)(-1/2)$.

Definition

- If $f(x) = f(-x)$ for all x in the domain of f, then f is an **even function**. An even function is symmetric about the y-axis.
- If $f(-x) = -f(x)$ for all x in the domain of f, then f is an **odd function**. An odd function is symmetric about the origin.



(a) Symmetry about the y-axis



(b) Symmetry about the origin

Example

Determine whether each of the following functions is even, odd, or neither.

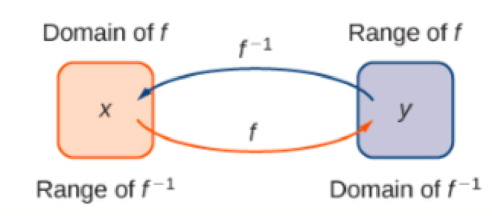
a. $f(x) = -5x^4 + 7x^2 - 2$

b. $f(x) = 2x^5 - 4x + 5$

c. $f(x) = \dfrac{3x}{x^2 + 1}$

Inverse Function

Given a function f with domain D and range R, its **inverse function** (if it exists) is the function f^{-1} with domain R and range D such that $f^{-1}(y) = x$ if $f(x) = y$. In other words, for a function f and its inverse f^{-1}, $f^{-1}(f(x)) = x$ for all x in D, and $f\left(f^{-1}(y)\right) = y$ for all y in R.



Problem-solving strategy: Finding an inverse function

- Solve the equation $y = f(x)$ for x.
- Interchange the variables x and y and write $y = f^{-1}(x)$.

Example

Find the inverse for the function $f(x) = 3x - 4$. State the domain and range of the inverse function. Verify that $f^{-1}(f(x)) = x$

1.1.6. Inverse functions

- The inverse function of sin, denoted "arcsin", and the inverse function of cos, denoted "arccos", are defined on the domain $D = \{x \mid -1 \leq x \leq 1\}$ as follows:

$$\begin{aligned} \arcsin(x) = y &\text{ if and only if } \sin(y) = x \text{ and } -\tfrac{\pi}{2} \leq y \leq \tfrac{\pi}{2} \\ \arccos(x) = y &\text{ if and only if } \cos(y) = x \text{ and } 0 \leq y \leq \pi \end{aligned}$$

- The inverse tangent function, denoted $\tan^{-1}$ or arctan, and inverse cotangent function, denoted $\cot^{-1}$ or arccot, are defined on the domain $D = \{x \mid -\infty < x < \infty\}$ as follows:

$$\begin{aligned} \arctan(x) = y &\text{ if and only if } \tan(y) = x \text{ and } -\tfrac{\pi}{2} < y < \tfrac{\pi}{2} \\ \operatorname{arccot}(x) = y &\text{ if and only if } \cot(y) = x \text{ and } 0 < y < \pi \end{aligned}$$

Exponential and logarithmic functions

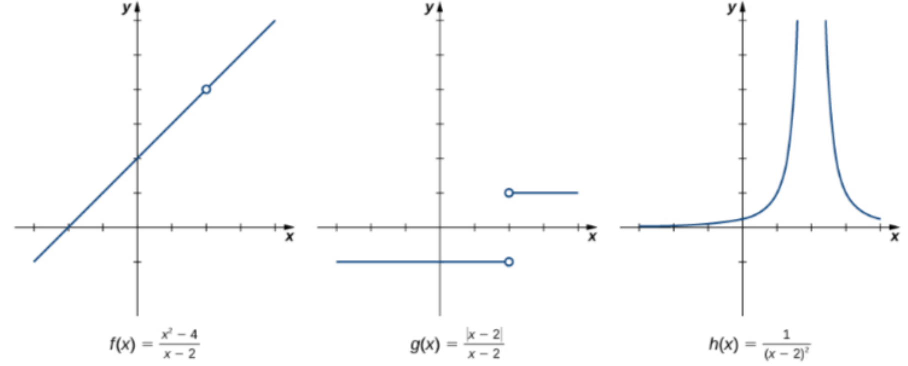
- The **exponential function** $y = b^x$ is increasing if $b > 1$ and decreasing if $0 < b < 1$. Its domain is $(-\infty, +\infty)$ and its range is $(0, \infty)$.
- The **logarithmic function** $y = \log_b(x)$ is the inverse of $y = b^x$. Its domain is $(0, \infty)$ and its range is $(-\infty, +\infty)$.
- The **natural exponential function** is $y = e^x$ and the natural logarithmic function is $y = \ln x = \log_e x$
- Given an exponential function or logarithmic function in base a, we can make a change of base to convert this function to any base $b > 0, b \neq 1$. We typically convert to base e.
- The **hyperbolic functions** involve combinations of the exponential functions e^x and e^{-x}. As a result, the inverse hyperbolic functions involve the natural logarithm.

Content

Look at the graphs of the functions

$$f(x) = \frac{x^2 - 4}{x - 2}, g(x) = \frac{|x - 2|}{x - 2}, \text{ and } h(x) = \frac{1}{(x - 2)^2}$$

which are shown in the following figure. In particular, let's focus our attention on the behavior of each graph at and around $x = 2$.



(a) (b) (c)

- Each of the three functions is undefined at $x = 2$, but if we make this statement and no other, we give a very incomplete picture of how each function behaves in the vicinity of $x = 2$.
- To express the behavior of each graph in the vicinity of 2 more completely, we need to introduce the concept of a limit.
- Let's first take a closer look at how the function $f(x) = \left(x^2 - 4\right) / (x - 2)$ behaves around $x = 2$. As the values of x approach 2 from either side of 2, the values of $y = f(x)$ approach 4. Mathematically, we say that the limit of $f(x)$ as x approaches 2 is 4. Symbolically, we express this limit as

$$\lim_{x \to 2} f(x) = 4$$

Definition

Let $f(x)$ be a function defined at all values in an open interval containing a, with the possible exception of a itself, and let L be a real number. If all values of the function $f(x)$ approach the real number L as the values of $x(\neq a)$ approach the number a, then we say that the **limit of** $f(x)$ **as** x **approaches** a **is** L. (More succinct, as x gets closer to a, $f(x)$ gets closer and stays close to L.) Symbolically, we express this idea as

$$\lim_{x \to a} f(x) = L$$

1.2.1. Definition

Recall that the distance between two points a and b on a number line is given by $|a - b|$

- The statement $|f(x) - L| < \varepsilon$ may be interpreted as: The distance between $f(x)$ and L is less than ε.
- The statement $0 < |x - a| < \delta$ may be interpreted as: $x \neq a$ and the distance between x and a is less than δ.
- The statement $|f(x) - L| < \varepsilon$ is equivalent to the statement $L - \varepsilon < f(x) < L + \varepsilon$
- The statement $0 < |x - a| < \delta$ is equivalent to the statement $a - \delta < x < a + \delta$ and $x \neq a$

1.2.1. Definition

Definition

Let $f(x)$ be defined for all $x \neq a$ over an open interval containing a. Let L be a real number. Then, the limit

$$\lim_{x \to a} f(x) = L$$

is defined by that, for every $\varepsilon > 0$ there exists a $\delta > 0$ such that

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

One-sided limits

- **Limit from the left**: Let $f(x)$ be a function defined at all values in an open interval of the form (c, a), and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x (where $x < a$) approach the number a, then we say that L is the limit of $f(x)$ as x approaches a from the left. Symbolically, we express this idea as $\lim_{x \to a^-} f(x) = L$
- **Limit from the right**: Let $f(x)$ be a function defined at all values in an open interval of the form (a, c), and let L be a real number. If the values of the function $f(x)$ approach the real number L as the values of x(where $x > a$) approach the number a, then we say that L is the limit of $f(x)$ as x approaches a from the right. Symbolically, we express this idea as $\lim_{x \to a^+} f(x) = L$

One-sided limits (continuity)

- **Limit from the right**: Let $f(x)$ be defined over an open interval of the form (a, b) where $a < b$. Then,

$$\lim_{x \to a^+} f(x) = L$$

  if for every $\varepsilon > 0$, there exists a $\delta > 0$ such that if $0 < x - a < \delta$, then $|f(x) - L| < \varepsilon$.

- **Limit from the left**: Let $f(x)$ be defined over an open interval of the form (b, a) where $b < a$. Then,

$$\lim_{x \to a^-} f(x) = L$$

  if for every $\varepsilon > 0$, there exists a number $\delta > 0$ such that if $-\delta < x - a < 0$, then $|f(x) - L| < \varepsilon$

Two types of one-sided limits

Example

For the function $f(x) = \begin{cases} x+1 & \text{if } x < 2 \\ x^2 - 4 & \text{if } x \geq 2 \end{cases}$, evaluate each of the following limits.

a. $\lim\limits_{x \to 2^-} f(x)$

b. $\lim\limits_{x \to 2^+} f(x)$.

Infinite limits

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a.

- If the values of $f(x)$ increase without bound as the values of x (where $x \neq a$) approach the number a, then we say that **the limit as x approaches a is positive infinity** and we write

$$\lim_{x \to a} f(x) = +\infty$$

- If the values of $f(x)$ decrease without bound as the values of x (where $x \neq a$) approach the number a, then we say that **the limit as x approaches a is negative infinity** and we write

$$\lim_{x \to a} f(x) = -\infty$$

Infinite limits

Let $f(x)$ be defined for all $x \neq a$ in an open interval containing a.

- We have an **infinite limit**

$$\lim_{x \to a} f(x) = +\infty$$

  if for every $M > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) > M$.

- We have a **negative infinite limit**

$$\lim_{x \to a} f(x) = -\infty$$

  if for every $M > 0$, there exists $\delta > 0$ such that if $0 < |x - a| < \delta$, then $f(x) < -M$.

Example

Evaluate each of the following limits, if possible. confirm your conclusion.

a. $\lim\limits_{x \to 0^-} \frac{1}{x}$

b. $\lim\limits_{x \to 0^+} \frac{1}{x}$

c. $\lim\limits_{x \to 0} \frac{1}{x}$

1.2.2. Operations of the limit

Let $f(x)$ and $g(x)$ be defined for all $x \neq a$ over some open interval containing a. Assume that L and M are real numbers such that $\lim\limits_{x\to a} f(x) = L$ and $\lim\limits_{x\to a} g(x) = M$. Let c be a constant. Then, each of the following statements holds:

- Sum law for limits:
  $\lim\limits_{x\to a}(f(x)+g(x)) = \lim\limits_{x\to a} f(x) + \lim\limits_{x\to a} g(x) = L + M$
- Difference law for limits:
  $\lim\limits_{x\to a}(f(x)-g(x)) = \lim\limits_{x\to a} f(x) - \lim\limits_{x\to a} g(x) = L - M$
- Constant multiple law for limits: $\lim\limits_{x\to a} cf(x) = c \cdot \lim\limits_{x\to a} f(x) = cL$
- Product law for limits:
  $\lim\limits_{x\to a}(f(x)\cdot g(x)) = \lim\limits_{x\to a} f(x) \cdot \lim\limits_{x\to a} g(x) = L \cdot M$
- Quotient law for limits: $\lim\limits_{x\to a} \frac{f(x)}{g(x)} = \frac{\lim\limits_{x\to a} f(x)}{\lim\limits_{x\to a} g(x)} = \frac{L}{M}$ for $M \neq 0$.
- Power law for limits: $\lim\limits_{x\to a}(f(x))^n = \left(\lim\limits_{x\to a} f(x)\right)^n = L^n$ for every positive integer n.

1.2.2. Operations of the limit

- Root law for limits: $\lim_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)} = \sqrt[n]{L}$ for all L if n is odd and for $L \geq 0$ if n is even and $f(x) \geq 0$.
- The Squeeze Theorem: Let $f(x), g(x)$, and $h(x)$ be defined for all $x \neq a$ over an open interval containing a. If

$$f(x) \leq g(x) \leq h(x)$$

for all $x \neq a$ in an open interval containing a and

$$\lim_{x\to a} f(x) = L = \lim_{x\to a} h(x)$$

where L is a real number, then $\lim_{x\to a} g(x) = L$.

1.2.2. Operations of the limit

Example

Use the limit laws to evaluate

a. $\lim\limits_{x \to -3} (4x + 2)$

b. $\lim\limits_{x \to 2} \frac{2x^2 - 3x + 1}{x^3 + 4}$

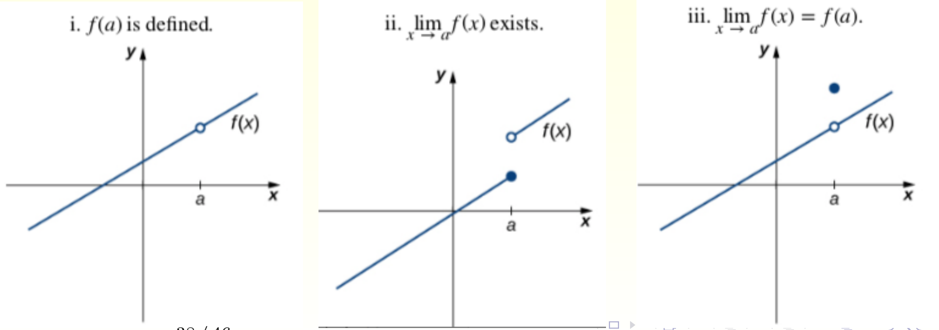
c. $\lim\limits_{x \to 6} (2x - 1)\sqrt{x + 4}$

d. $\lim\limits_{x \to 0} x \cos x$.

Content

1.3.1. Continuity at a point

A function $f(x)$ is **continuous** at a point a if and only if the following three conditions are satisfied:
i. $f(a)$ is defined
ii. $\lim\limits_{x \to a} f(x)$ exists
iii. $\lim\limits_{x \to a} f(x) = f(a)$
A function is **discontinuous** at a if it fails to be continuous at a.



Example

Using the definition, determine whether the function

a. $f(x) = \left(x^2 - 4\right)/(x - 2)$ is continuous at $x = 2$

b. $f(x) = \begin{cases} -x^2 + 4 & \text{if } x \leq 3 \\ 4x - 8 & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$.

Example

Find the constant c that makes g continuous on $(-\infty, \infty)$.

$$g(x) = \begin{cases} x^2 - c^2 & \text{if } x < 4 \\ cx + 20 & \text{if } x \geqslant 4 \end{cases}$$

Continuity from the right and from the left

- A function $f(x)$ is said to be **continuous from the right** at a if $\lim\limits_{x \to a^+} f(x) = f(a)$.
- A function $f(x)$ is said to be **continuous from the left** at a if $\lim\limits_{x \to a^-} f(x) = f(a)$.

Example

Find the interval(s) that the function $f(x) = \frac{x-1}{x^2+2x}$ is continuous.

Continuous on an interval

A function f is **continuous on an interval** if it is continuous at every point in the interval. (If f is defined only on one side of an endpoint of the interval, we understand continuous at the endpoint to mean continuous from the right or continuous from the left.)

Example

Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$

1.3.3. The properties of continuous functions

Theorem

- If f and g are continuous at x_0 and c is a constant, then the following functions are also continuous at x_0 : $f+g$, $f-g$, cf, fg, and $\dfrac{f}{g}$ if $g(x_0) \neq 0$.
- Any polynomial is continuous everywhere; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$
- Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.
- The following types of functions are continuous at every number in their domains: polynomials, rational functions, root functions, trigonometric functions.

The properties of continuous functions

- If f is continuous at b and $\lim_{x\to a} g(x) = b$, then $\lim_{x\to a} f(g(x)) = f(b)$. In other words,

$$\lim_{x\to a} f(g(x)) = f\left(\lim_{x\to a} g(x)\right).$$

- If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a.

The properties of continuous functions (cont.)

On what intervals is each function continuous?

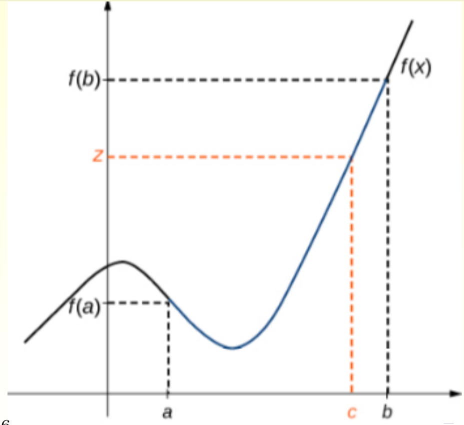
(a) $f(x) = x^{100} - 2x^{37} + 75$

(b) $g(x) = \dfrac{x^2 + 2x + 17}{x^2 - 1}$

(c) $h(x) = \sqrt{x} + \dfrac{x+1}{x-1} - \dfrac{x+1}{x^2+1}$

The Intermediate Value Theorem

Let f be continuous over a closed, bounded interval $[a, b]$. If z is any real number between $f(a)$ and $f(b)$, then there is a number c in $[a, b]$ satisfying $f(c) = z$



Example

- Show that there is a root of the equation $4x^3 - 6x^2 + 3x - 2 = 0$ between 1 and 2 .
- Show that $f(x) = x^3 - x^2 - 3x + 1$ has at least one root.